CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 14: Equation of a Line

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the slope of the line through $(-1, 4)$ and $(3, -4)$.
2. Find the slope of a line whose inclination is 60°, and of a line whose inclination is 30°.
3. Find k if the point $(3, k)$ lies on the line $2x - 5y = 1$.
4. Find the equation of the line with slope -3 and y-intercept 4.
5. Find the equation of the line through $(-2, 1)$ and $(4, 4)$.
6. Find the slope and the y-intercept of the line $5x - 2y + 6 = 0$.
7. Find the equation of the line through $(1, -2)$ parallel to the line $y = 4x + 7$.
8. Find the equation of the line through $(3, 2)$ perpendicular to the line $x + 2y = 6$.
9. Find the value of p if the lines $2x - py + 3 = 0$ and $4x + 2y - 5 = 0$ are perpendicular.
10. Find the equation of the perpendicular bisector of the segment joining $(1, 5)$ and $(7, 1)$.

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Answer Key

1. -2
2. $\sqrt{3}$ and $\frac{1}{\sqrt{3}}$
3. $k = 1$
4. $y = -3x + 4$
5. Slope $\frac{1}{2}$; $x - 2y + 4 = 0$
6. Slope $\frac{5}{2}$; y-intercept 3
7. $y = 4x - 6$
8. Slope 2; $y = 2x - 4$
9. $\frac{2}{p} \times (-2) = -1$, so $p = 4$
10. Mid-point $(4, 3)$, slope $\frac{3}{2}$; $3x - 2y - 6 = 0$

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